

Rahul Rachhoya

AI ENGINEER | LLM AGENTS, RAG, FINE-TUNING & LLM SERVING

Gurugram, India | +91-8386067676 | its.rahul.rachhoya@gmail.com
linkedin.com/in/rahul-rachhoya | github.com/RahulRachhoya

PROFILE

AI engineer with 4+ years in software engineering and 3+ years in applied AI: RAG with hybrid search and reranking (retrieval **85% to 91%**), LangGraph agents and AWS deployments at **99.7% uptime**. Four public projects go below the API layer: QLoRA fine-tuning, vLLM serving on Kubernetes, retrieval benchmarks and a deployed TypeScript agent. Available immediately, remote from India.

TECHNICAL SKILLS

LLMs & agents: LangGraph, LangChain, LlamaIndex, Vercel AI SDK, function calling, Claude, GPT-4, AWS Bedrock
Fine-tuning & serving: QLoRA/LoRA, PEFT, TRL, Hugging Face Transformers, PyTorch, scikit-learn, vLLM, Kubernetes
Retrieval & data: hybrid BM25 + dense search, RRF, Cohere reranking, Qdrant, pgvector, PostgreSQL, MongoDB
Engineering & cloud: Python, TypeScript, Next.js, Node.js, FastAPI, Docker, AWS Lambda, EC2, S3, Vercel, pytest
Evaluation & MLOps: LangSmith, Weights & Biases, MLflow, Prometheus, nDCG/Recall evals, A/B testing

PROFESSIONAL EXPERIENCE

Careers360

AI Engineer - Voice AI & Conversational Systems

Hyderabad, India

Jan 2026 - Sep 2026

- Owned a production voice AI career counselor using Claude on AWS Bedrock, Sarvam AI STT/TTS and pgvector retrieval, serving **10K+ sessions**; RAG grounding reduced hallucinations by 35%.
- Integrated LLM function calling into multi-agent workflows, improving accuracy by 28%; deployed services on AWS Lambda with **99.7% uptime**.
- Reduced per-session cost by **40%**, from **\$0.40 to \$0.24**, through prompt caching and A/B testing; tracked latency, cost and quality with LangSmith.

Crystaltech Services

AI Engineer

Remote, India

Aug 2024 - Dec 2025

- Built a four-agent document workflow with LangGraph, processing 500+ documents at **94% accuracy** and reducing handling time from **30 minutes to 7 minutes**.
- Improved retrieval from **85% to 91%** with BM25 + vector search and Cohere reranking; A/B tested GPT-4 and Claude configurations across 50+ tracked iterations for 60% cost savings.

STL Digital Limited

System Engineer (AI/ML Focus)

Pune, India

Jun 2022 - Jul 2024

- Implemented a ChromaDB and GPT-3.5 RAG system that improved accuracy from **65% to 91%**, serving 500+ daily queries at a 1.8-second response time; maintained 85% test coverage with pytest and GitHub Actions CI/CD.

PROJECTS

- finetune-intent-lab** (github.com/RahulRachhoya/finetune-intent-lab): QLoRA fine-tune of Qwen2.5-1.5B for 77-class intent classification, **36.9% to 93.4%** accuracy training 1.2% of the parameters; beat TF-IDF + logistic regression (91.1%), PyTorch (90.0%), tracked in MLflow.
- llm-serving-k8s** (github.com/RahulRachhoya/llm-serving-k8s): served that adapter with vLLM on a GPU Kubernetes cluster (k3d): **185.5 req/s** at 32 in flight, 78 ms p50 for a single request, 93.2% served accuracy (93.4% at training time).
- agent-memory-lab** (github.com/RahulRachhoya/agent-memory-lab): Qdrant hybrid vs pgvector on BEIR SciFact (nDCG@10 **0.725** vs 0.713 dense); LangGraph agent on Claude with MongoDB memory and telemetry.
- ai-sdk-research-agent** (github.com/RahulRachhoya/ai-sdk-research-agent, live: ai-sdk-research-agent.vercel.app): TypeScript agent on Next.js and the Vercel AI SDK; matched Python BM25 token IDs on 300/300 queries, found the Qdrant JS client **20x slower** than node:http; deployed on Vercel and Render.

EDUCATION & CERTIFICATION

Master of Computer Applications (MCA), University of Hyderabad, India | GPA: 8.2/10

2019 - 2022

Certification: Deep Learning Specialization, Coursera - completed.